

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Semestertoets 2
Kopiereg voorbehou

Semester Test 2
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Module EBN111
Elektrisiteit en Elektronika
22 April 2016

Module EBN111
Electricity and Electronics
22 April 2016



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Exam information:

Eksamensinligting:

Maximum marks: Maksimum punte:	45	Full marks: Volpunte:	46
Duration of paper: Duur van vraestel:	90 minutes 90 minute	Open / closed book: Oopboek / toeboek:	Closed Toe
Total number of pages (including this page): Totale aantal bladsye (hierdie blad ingesluit):	16		

IMPORTANT - BELANGRIK

- The examination regulations of the University of Pretoria apply.*
Die eksamenregulasies van die Universiteit van Pretoria geld.
- All questions must be answered on the ANSWER SHEET.*
Alle vrae moet op die ANTWOORDBLAD beantwoord word.
- Students may do their own rough calculations on the question paper – the question paper will not be taken in.*
Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
- Answers must include units!*
Antwoorde moet eenhede insluit!
- Final answers must be written as real numbers, and not as fractions.*
Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.

Dosent(e): Lecturer(s):	1. G Hancke
	2. D Bhatt
	3. D le Roux

VRAAG/QUESTION 1 [13]

Question 1.1 [4]

Use mesh analysis to determine the coefficients a , b , c and d in the following equations:

$$ai_1 + bi_3 = 7$$

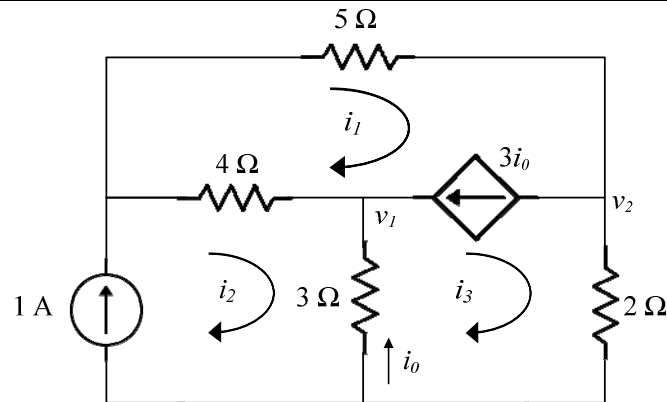
$$ci_1 + di_3 = 3$$

Vraag 1.1 [4]

Gebruik lus-analise om die koëffisiente a , b , c en d in die volgende vergelykings te bepaal:

$$ai_1 + bi_3 = 7$$

$$ci_1 + di_3 = 3$$

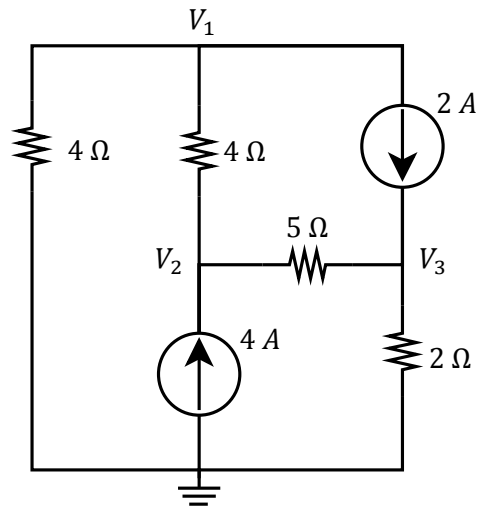


Vraag 1.2 [3]

Stel 'n matriksvergelyking d.m.v. inspeksie op in terme van die node spannings.

Question 1.2 [3]

Use inspection to set up a matrix equation in terms of the nodal voltages.



$$\begin{bmatrix} 0.5 & & \\ & & -0.2 \\ & & \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} \\ \\ 2 \end{bmatrix}$$

Question 1.3 [4]

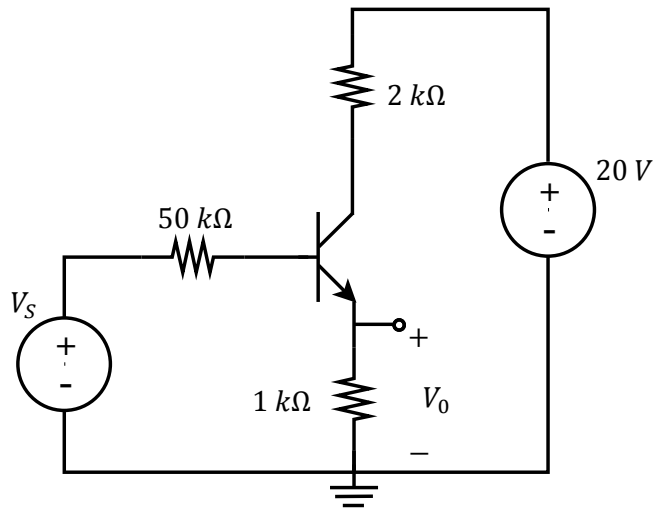
Given: $\beta = 99$ and $V_{BE} = 0.7$ V.

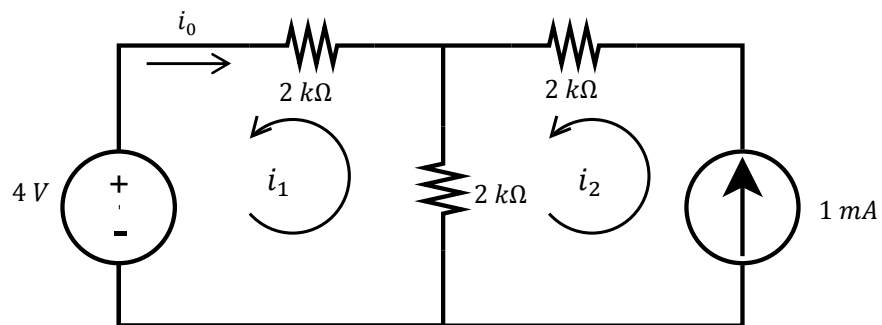
- a) If $V_0 = 4$ V, determine V_S . [2]
b) If $I_B = 20$ μ A, determine V_{CE} . [2]

Vraag 1.3 [4]

Gegee: $\beta = 99$ en $V_{BE} = 0.7$ V.

- a) Indien $V_0 = 4$ V, bepaal V_S . [2]
b) Indien $I_B = 20$ μ A, bepaal V_{CE} . [2]



Question 1.4 [2]Use mesh analysis and solve for i_0 .**Vraag 1.4 [2]**Gebruik lusanalyse en los op vir i_0 .

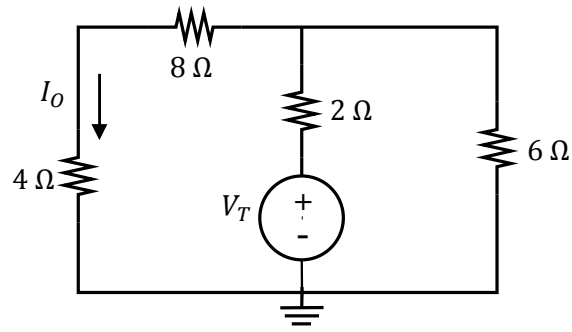
VRAAG/QUESTION 2 [22]

Question 2.1 [2]

If $I_0 = 1\text{ A}$, what is V_T ?

Vraag 2.1 [2]

Indien $I_0 = 1\text{ A}$, wat is V_T ?



Question 2.2 [4]

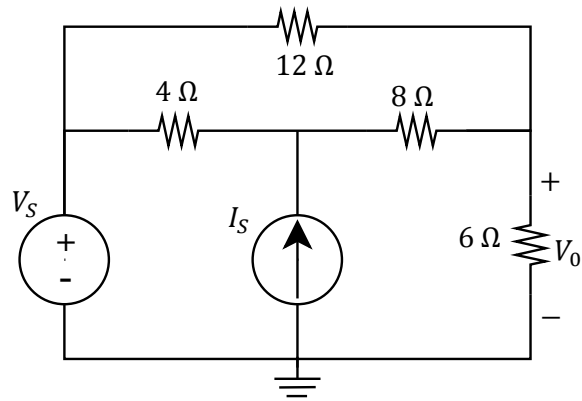
The voltage source V_S contributes 10 V to V_0 , and the current source I_S contributes 4 V to V_0 . Therefore, the total is $V_0 = 10 + 4 = 14$ V.

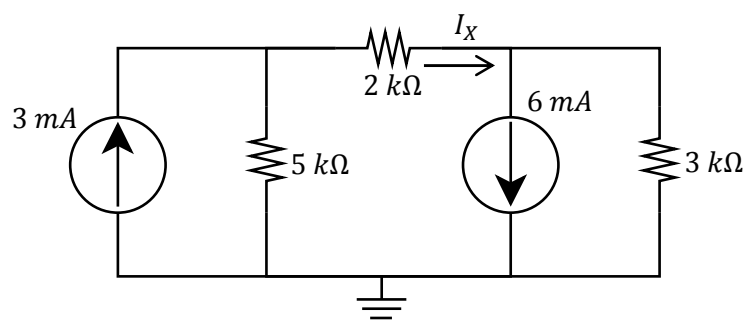
Determine V_S and I_S .

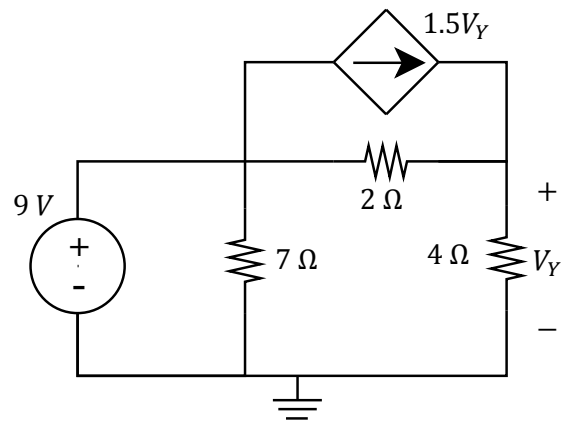
Vraag 2.2 [4]

Die spanningsbron V_S dra 10 V by tot V_0 , en die stroombron I_S dra 4 V by tot V_0 . Die totaal is dus $V_0 = 10 + 4 = 14$ V.

Bepaal V_S en I_S .



Question 2.3 [2]Make use of source transformation and determine I_X .**Vraag 2.3 [2]**Maak gebruik van brontransformatie en bepaal I_X .

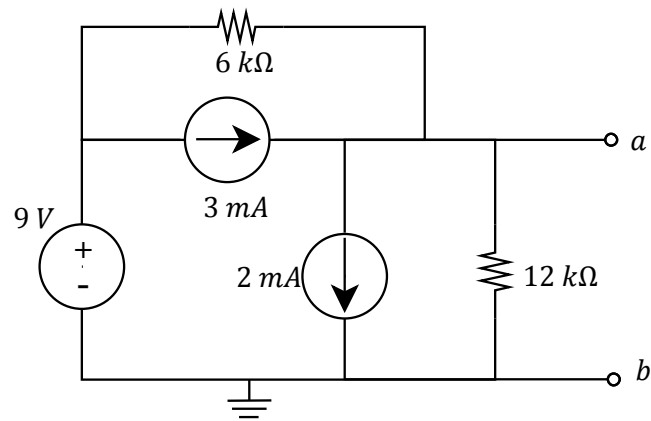
Question 2.4 [2]Make use of source transformation and determine V_Y .**Vraag 2.4 [2]**Maak gebruik van brontransformatie en bepaal V_Y .

Question 2.5 [4]

Determine the Thevenin equivalent voltage V_{TH} and resistance R_{TH} w.r.t. terminals a - b .

Vraag 2.5 [4]

Bepaal die Thevenin ekwivalente spanning V_{TH} en weerstand R_{TH} t.o.v. terminale a - b .

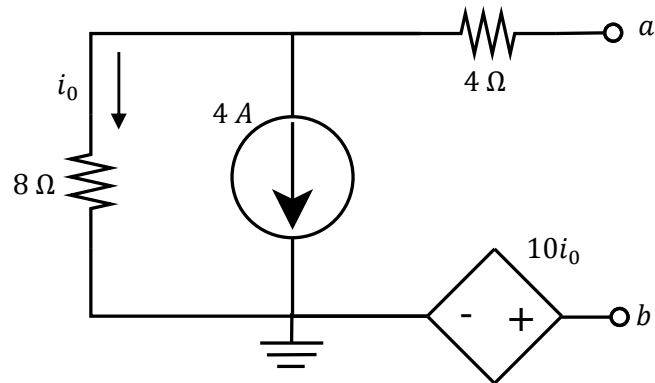


Question 2.6 [4]

Determine the Norton equivalent current I_N and resistance R_N wrt terminals a - b .

Vraag 2.6 [4]

Bepaal die Norton ekwivalente stroom I_N en weerstand R_N tov terminale a - b .



Question 2.7 [4]

- a) The Norton equivalent of a circuit is $I_N = 10 \text{ mA}$ and $R_N = 5 \text{ k}\Omega$. What is the maximum power P_{max} that can be delivered to a load resistor?
- b) The Thevenin equivalent of a circuit is $R_{TH} = 8 \text{ }\Omega$ and $V_{TH} = 5 \text{ V}$. If a load resistance of $2 \text{ }\Omega$ is connected to the circuit, what is the power P_L absorbed by the load?

Vraag 2.7 [4]

- a) Die Norton ekwivalent van 'n kringbaan is $I_N = 10 \text{ mA}$ en $R_N = 5 \text{ k}\Omega$. Wat is die maksimum drywing P_{max} wat aan 'n lasweerstand oorgedra kan word?
- b) Die Thevenin ekwivalent vir 'n kringbaan is $R_{TH} = 8 \text{ }\Omega$ en $V_{TH} = 5 \text{ V}$. Indien 'n lasweerstand van $2 \text{ }\Omega$ aan die kringbaan gekoppel word, wat is die drywingsverbruik P_L van die lasweerstand?

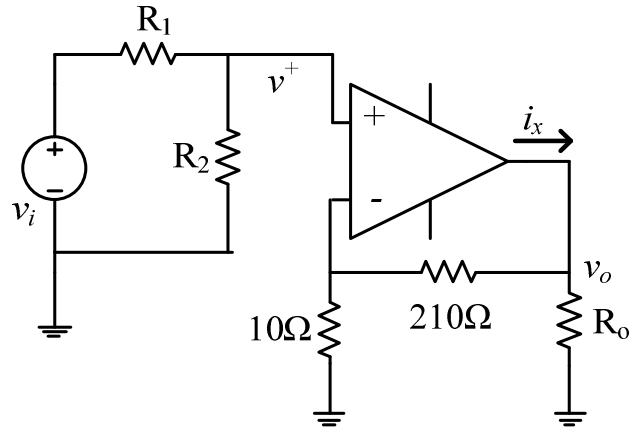
QUESTION/ VRAAG 3 [11]

Question 3.1 [4]

- a) Calculate the closed-loop gain (v_o/v_i) of the circuit when $R_1=R_2$
- b) Find the current i_x when $R_o = 110\ \Omega$ and $v^+ = -2\text{mV}$.

Vraag 3.1 [4]

- a) Bereken die geslote-lus wins (v_o/v_i) van die kringbaan indien $R_1 = R_2$.
- b) Bereken die stroom i_x indien $R_o = 110\ \Omega$ en $v^+ = -2\text{mV}$



Question 3.2 [3]

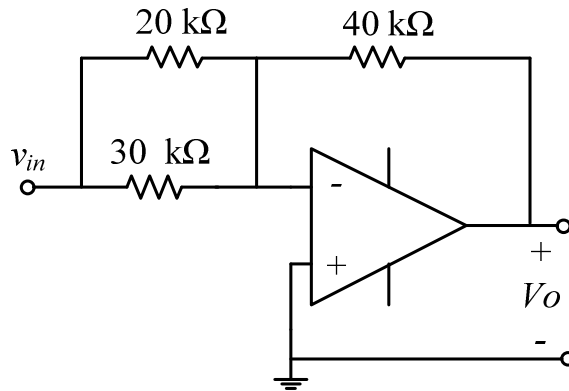
For the circuit below:

- identify the type of amplification circuit,
(*Non-Inverting Amplifier, Inverting Amplifier, Buffer, Summing Amplifier, or None of the above.*)
- calculate the output voltage V_o , if $v_{in} = 15\text{mV}$.

Vraag 3.2 [3]

Vir onderstaande kringbaan:

- identifiseer die tipe versterker kringbaan,
(*Nie-Omkeer Versterker, Omkeer Versterker, Buffer, Sommeerder, of Nie een van bogenoemde.*)
- bereken die uitset spanning V_o , indien $v_{in} = 15\text{mV}$.

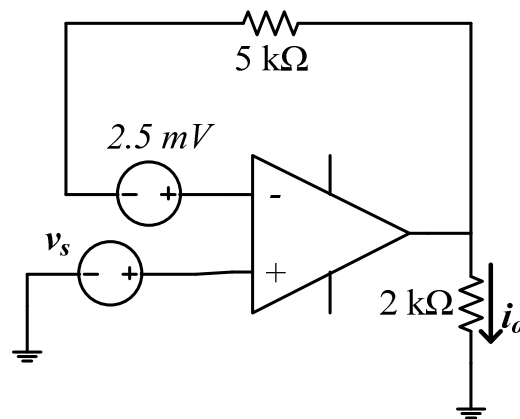


Question 3.3 [2]

What value of v_s will cause a current of $i_o = 4\ \mu\text{A}$ to flow through the $2\text{k}\Omega$ resistor?

Vraag 3.3 [2]

Vir watter waarde van v_s sal 'n stroom van $i_o = 4\ \mu\text{A}$ deur die $2\text{k}\Omega$ resistor vloei?



Question 3.4 [2]

If $v_s = 1 \text{ mV}$, calculate the output voltage v_o .

Vraag 3.4 [2]

Indien $v_s = 1 \text{ mV}$, bereken die uitset spanning v_o .

